Problem 1: Jowulu (17 marks)

Here are some arithmetic equations in Jowulu[1]. All numbers are less than 200.

fulpʊn × fulpʊn	=	yaa bʒei vapʃire
tɛ̱mpʊn × fuuli	=	bvʊnfulpʊn
bvʊnfulpʊn + bʒei	=	kɔn vatɛ̱ɛ̱
bʒei × fuuli	=	tontɛ̱ɛ̱
pʃire + pʃire	=	fulpʊn
fuuli$^{\text{tɛ̱ɛ̱na}}$	=	fuuli
bvʊntɛ̱ɛ̱ × tɛ̱mpʊn	=	yaa pʃire vabvʊntɛ̱mpʊn
ta̱a̱^{bʒei}	=	yaa tontɛ̱ɛ̱ vata̱a̱

Assignment 1: (8 marks)

Rewrite the equations above with regular (Hindu-Arabic) numerals (like “1 + 1 = 2”).

Assignment 2: (6 marks)

Write in Jowulu: (a) 78; (b) 185; (c) 25.

Assignment 3: (3 marks)

Write with regular (Hindu-Arabic) numerals:

(a) yaa tontɛ̱ɛ̱ vabvʊntɛ̱mpʊn

(b) yaa pʃire vata̱a̱

– Animikha Dutta Dhar

Problem 2: GaeilGo Doesn’t Go (20 marks)

Everybody and everything in Ireland warmly welcomes visitors. Well, almost everything, as Aalok finds out when he decides to visit; it seems impossible to figure out the pronunciation of Gaeilge[2] words from their spelling! Aalok turns to his trusty pronunciation app called “GaeilGo” for help.

Given below are some Gaeilge words and their pronunciations. The pronunciations use the International Phonetic Alphabet (IPA), but knowledge of IPA is not required to solve this problem, except that the ˈ mark before a syllable means the syllable is stressed.

[1]Jowulu, sometimes also referred to as Jɔ or Samogho, is a language of the Mande family spoken in Mali and Burkina Faso by around 10,000 to 15,000 people.

[2]“Gaeilge” is the Irish word for “Irish”, the native language of Ireland.

Gaeilge word	Pronunciation	Gaeilge word	Pronunciation
scuab	/sʸkuəbʸ/	fluairít	/fʸlʸuəˈɾʲiːtʲ/
scéimh	/ʃceːvʲ/	nialas	/ˈn̪ʲiəlʸəsʸ/
bliain	/bʲlʲiənʲ/	toscaire	/ˈtʸɔsʸkəɾʲə/
leaid	/lʲadʲ/	tacar	/ˈtʸakəɾʸ/
gocarsach	/ˈgʸɔkəɾʸsʸəx/	deoire	/ˈdʲoːɾʲə/
neachtar	/n̪ʲaxtʸəɾʸ/	neodrú	/ˈn̪ʲoːdʸɾʸuː/
geafáil	/ˈɟafʸaːlʲ/	gleo	/ɟlʲoː/
meacan	/ˈmʲakənʸ/	bádchlós	/ˈbʸaːdʸxlʸoːsʸ/
tábhacht	/ˈtʸaːvʸəxtʸ/	bibe	/ˈbʲɪbʲə/
limfe	/ˈlʲɪmʲfʲə/	treoluas	/ˈtʲɾʲoːlʸuəsʸ/
latrach	/ˈlʸatʸɾʸəx/	dobrón	/dʸɔbʸˈɾʸoːnʸ/
sintéisach	/ʃɪnʲˈtʲeːʃəx/	meaintréimhse	/mʲaːnʲˈtʲɾʲeːvʲʃə/
chéad	/çeːdʸ/	feoil	/fʲoːlʲ/
deaide	/ˈdʲadʲə/	frigiach	/ˈfʲɾʲɪgiəx/

Assignment 1: (13 marks)

Aalok's phone is running out of battery—but perhaps he doesn't need it anymore. How should the following Gaeilge words be pronounced? In each word, the first syllable is stressed, unless an underline indicates a different stressed syllable.

Gaeilge word	Pronunciation	Gaeilge word	Pronunciation
béarla	_____ (a) _____	móráireamh	_____ (h) _____
riachtanais	_____ (b) _____	ordóige	_____ (i) _____
spiaire	_____ (c) _____	moltóirí	_____ (j) _____
débhliantúil	_____ (d) _____	deoiseach	_____ (k) _____
luachmhar	_____ (e) _____	giaráilte	_____ (l) _____
eolach	_____ (f) _____	buaibheach	_____(m)_____
meaitseáil	_____ (g) _____		

Assignment 2: (7 marks)

Explain your solution.

– Rujul Gandhi and Shardul Chiplunkar